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An Analysis of Austin's Housing Market
Description of the Project

For this study, our team decided to investigate the Austin, Texas House Listings dataset, aiming to analyze shifts in the housing market dynamics over recent years. Utilizing this dataset, our objective was to identify potential correlations between distinct house attributes and their corresponding prices across various zip codes in the Austin, Texas area. Moreover, the dataset included more nuanced factors such as the average school ratings to allow our team to gain a comprehensive understanding.

Our initial interest in this dataset was piqued by its direct relevance to us as we are Austin residents, who have seen a sharp increase in housing prices, especially in the West Campus rental market. With Austin's status as a rapidly growing urban city, we realize that housing prices are one of people's top concerns when thinking about moving to Austin. Therefore, our analysis aims to offer insight into the Austin housing market, serving to help individuals make informed decisions about where to live and what factors to consider.

Exploratory Analysis

Upon analyzing the distribution of home sales across various zip codes, as illustrated in Figure 1.1 - 1.4, a notable trend emerged: a significant portion of the homes sold concentrated in South Austin. This observation suggests that South Austin may possess attributes desirable to buyers seeking a blend of trendiness and affordability. The area's appeal could stem from factors such as proximity to urban amenities, vibrant cultural scene, and comparatively reasonable housing prices.

We also decided to look at the correlation between school-related and housing factors. Due to little to no correlation, we looked at correlations between the factors pertaining to schools. Figure 2 shows a strong positive correlation coefficient of 0.673690 between 'avgSchoolRating' and 'MedianStudentsPerTeacher'. This suggests that as the average school rating increases, the median number of students per teacher also tends to increase, contradicting the conventional rule that a lower student-teacher ratio allows students to perform better due to more personalized attention.

In addition, we examined the number of houses constructed over the years. Figure 3 reveals a notable decrease in homes built between 1984 and 1990, mirroring the 'worst housing crash' in Austin from 1986-1994 (Gerli,

2021). We also found a significant drop in construction in 1999, 2006, and 2017. These construction drops appear to happen 2-4 years before major economic crashes - Dot Com, 2008 housing crash, COVID 19. We do not have a definitive reason why this happens, but this would be a good area to study forward if the crashes are not coincidental.

In our exploration of size features, we found that average living area, number of bedrooms, and number of stories were highly correlated across decades (Figure 4). Lot size (SQFT) appeared to have a significant increase in the 1950's, but when we exclude outliers in this period (Figure 5), the average 'lotSize' for the decade becomes on par with other decades.

We also looked into features that influenced house price volatility by looking into the number of price changes for a listing. Running the same models, we discovered that there were no statistically significant features that could help predict housing price changes, including those based on location, as the R^2 calculated from the models was ~0.04. There were also no significant trends between region and price changes(Figure 6). However, further research suggests that zipcodes with higher prices and number of price changes on average tend to be newly developed areas.

Building the Model

Looking through the data we discovered that cleaning was necessary. While we did not find many NaNs or inf values, the cheapest house was listed at $5,500. Searching up the address led us to discover that the location was a lakefront property with 5 beds and 5 baths, and cost over 1.4 million dollars (Figure 7). Many of the cheapest houses were similarly incorrect. We set an arbitrary cutoff for the lower bound at $100,990, which is when the housing prices became similar to their listings on Zillow. Austin also has some extremely expensive houses, so we set an upper bound to anything above 1.5 times the interquartile range above the 75th quartile.

We did some feature engineering to create features we thought may have been important, though only 2 of them (DecadeBuilt and storiesLivingRatio) ended up being an important feature. Also, storiesLivingRatio was the only engineered feature our final model (Random Forest) found was important.

We created 3 models: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. For Random Forest Regression and Gradient Boosting we used the GridSearch library to tune the hyperparameters to determine the optimal values for a given model, as shown in Figure 8. We also calculated the MSE and R^2 for each of the

models. As you can see in the figure, Random Forest and Gradient Boosting Regressors far outperformed the Linear Regression. Linear Regression had a negative R^2, which likely means it was overfit to the training set, and performed worse than picking the average price. Of all the models, Random Forest performed the best.

Solution & Insight

Distribution of Price Among Zip Codes

Examining the basic statistics for this cleaned dataset in Figure 9.2, we gained valuable insights into the housing market in Austin, Texas. It's important to note that before cleaning, the dataset was drastically different, as seen in Figure 9.1 After cleaning the data, a total of 14,038 homes were sold, with the dataset's mean price standing at $427,570 and a standard deviation of $172,580. What this indicates is that housing prices in Austin are generally more expensive than other cities, a trend that's supported by an article about average home prices across cities in Texas (Ramsey Solutions, 2024). Additionally, the quartiles of the clean dataset reveal that 25% of house prices fall below $299,999 suggesting that a significant portion of homes sold are reasonably priced. In addition, 75% of housing prices fall below $524,999, which showcases the upper range of housing prices in Austin, Texas.

Delving deeper into the statistics, when viewing the standard deviations of the house prices across zip codes, there were generally smaller house price standard deviations in zip codes East of Austin, while the house prices in West Austin had larger standard deviations (Figure 12). What was concluded was that due to the higher prices of homes in West Austin, the standard deviations were going to be proportionately larger than when compared to homes in East Austin. It also indicates that in East Austin there is less volatility, which can be beneficial for those shopping for a home. What this means for West Austin is that prices are more volatile, which can indicate more risk for those that are looking for a home.

Another focal point of our analysis was the distribution of price among different zip codes in Austin, Texas. By grouping the data by zip code and calculating the average prices, we were able to identify the five most and least expensive zip codes. Figure 1.1 illustrates our findings, indicating that the priciest zip codes are predominantly located in West Austin, particularly along the Colorado River (Figure 1.2). Conversely, the five least expensive zip codes are primarily situated in East Austin, as depicted in Figure 1.4. This could be due to a farther distance from downtown or lack of a riverside view. We also found that the zipcode we live in, 78705, is ranked as the 10th most variable zipcode and has the 14th highest mean (Figure 10).

Feature Importance

Both Random Forest and Gradient Boosting included the same eight features in their top 10 features with only two not included in both (Figure 11). They both included Living Area (SQFT), Latitude & Longitude, Average School Distance, Year Built, Lot Size (SQFT), Number of Price Changes, and Average School Distance. For the two features that they differed in, only Random Forest ranked stories living ratio and only gradient boosting ranked median students per teacher. Of all the engineered features, only the stories-to-living ratio was ranked. Between Random Forest and Gradient Boosting, the eight features that were found in both were ranked very similarly.

For both models, housing features were not ranked as favorably which is somewhat surprising since it seems that housing features can play a major role in influencing the price. For example, the bottom 10 features had some interesting results, especially since the Number of Waterfront Features and the presence of a View were deemed unimportant in the model (Figure 11). This went against our thought that the homes along the Colorado River were more expensive due to their proximity to the river. This dataset is composed of houses that are mostly similar to each other in the features they contain since the majority of them are of the same home type (single family homes). More emphasis was placed on features dependent on the general environment of an area such as school quality, price volatility, and longitude/latitude. General housing characteristics that are more distinguishing like lot size, living area and year built were also ranked.

Going off of Random Forest which had the highest accuracy, it seems that education-related features (school rating, school size, etc.) have the most influence on the latest housing price. This is unsurprising as it is common for more affluent neighborhoods to have higher quality education systems due to multiple factors. Factors such as property taxes, rated highly by linear regression, rise with housing prices which are directly associated with school expenditures, so more affluent areas will tend to have more available funding for their local schools. The association of housing prices and education has been shown to create a positive feedback loop that drives inequality as lower family households are further barred from accessing quality education. With further analysis into housing prices in Austin and additional socioeconomic and location data, we can look more into patterns of inequality.

Figure 1.1 - Top 5 Most Expensive Zip Codes

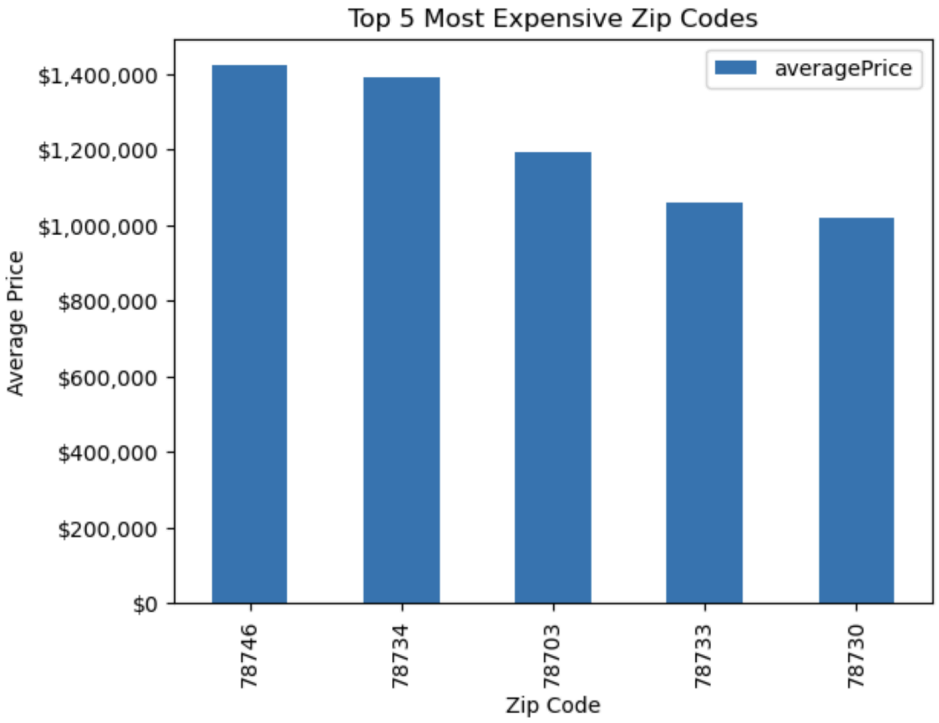


Figure 1.2 - Austin, Texas Zip Code 78746

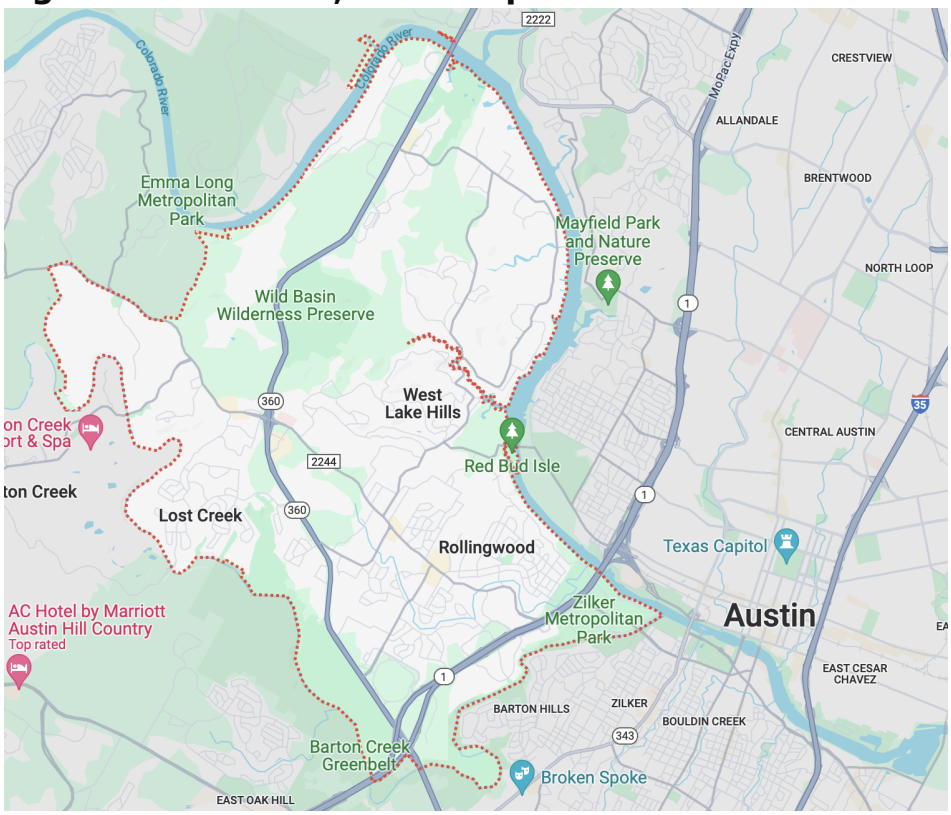


Figure 1.3 - Top 5 Least Expensive Zip Codes

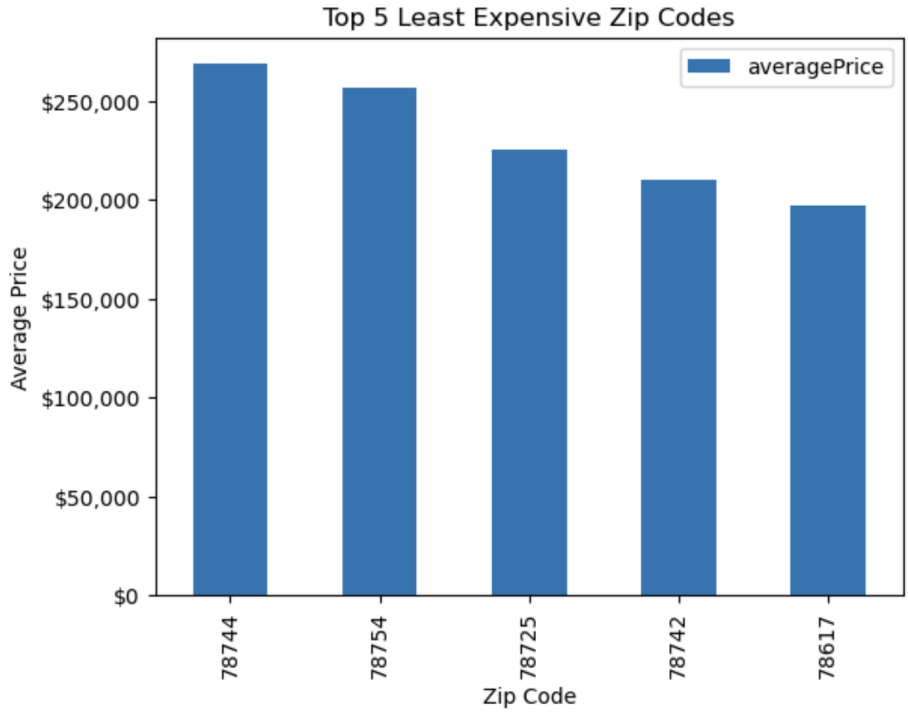


Figure 1.4 - Austin, Texas Zip Code 78617

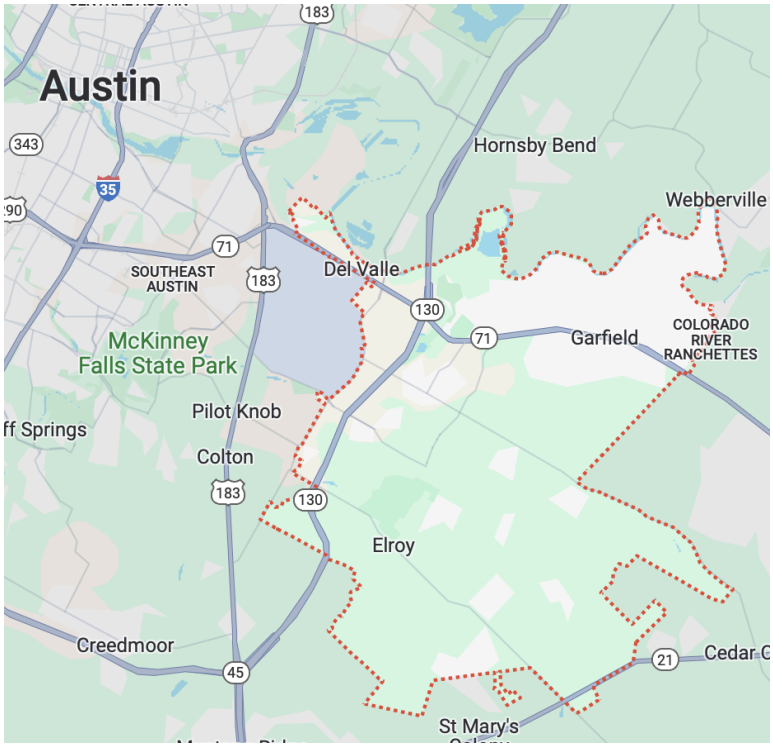


Figure 2 - Correlation coefficients of School-related factors

	MedianStudentsPerTeacher
numOfPrimarySchools	0.017797
numOfElementarySchools	-0.025148
numOfMiddleSchools	-0.105488
numOfHighSchools	-0.236477
avgSchoolDistance	0.160046
avgSchoolRating	0.673690
avgSchoolSize	0.399676
MedianStudentsPerTeacher	1.000000

Figure 3 - Distribution of Construction

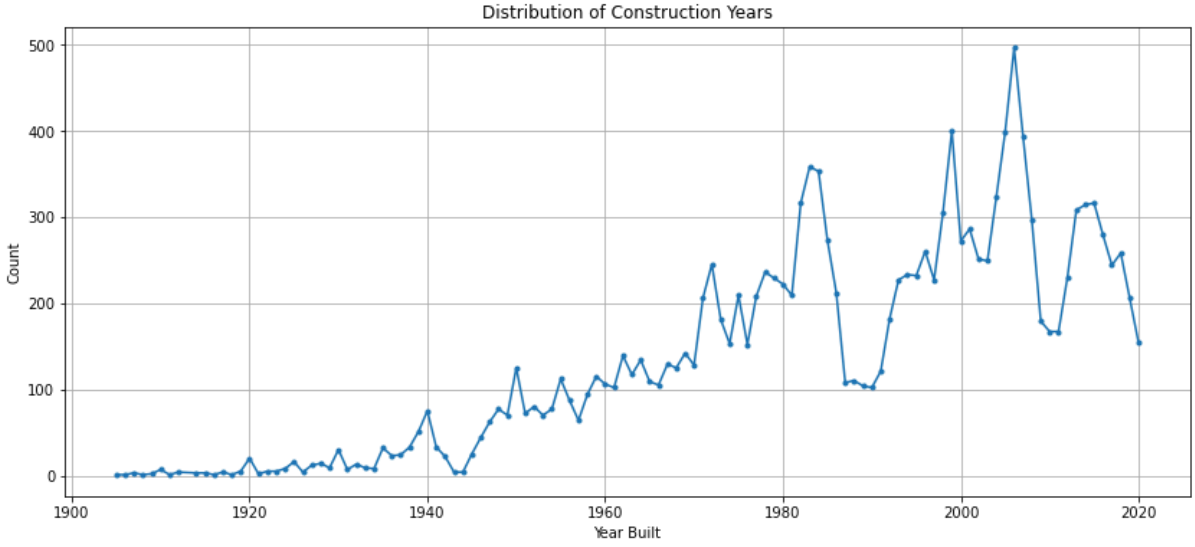


Figure 4 - House Size Characteristics by Decade of Construction

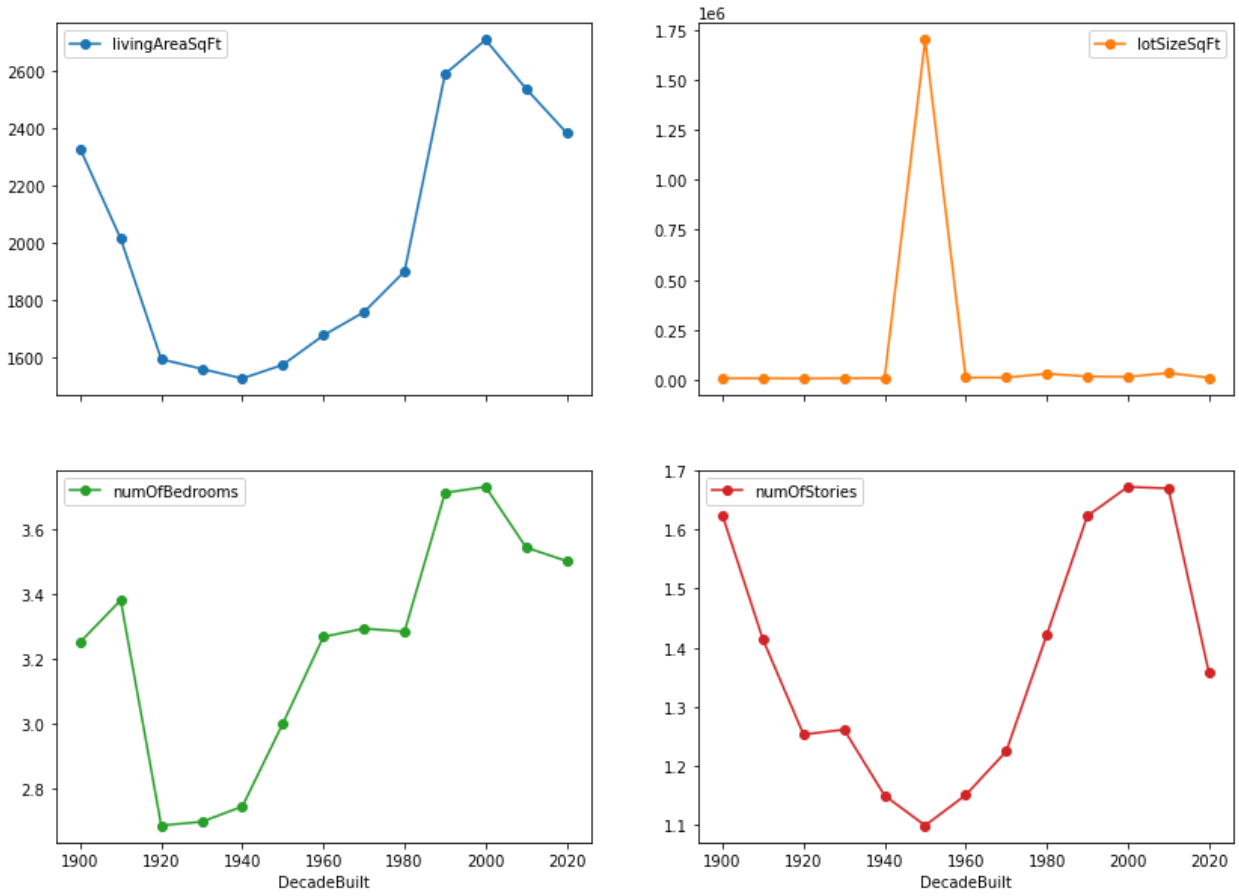


Figure 5 - Lot Size Outlier

```
df[['lotSizeSqFt']].describe()
```

	lotSizeSqFt
count	1.517100e+04
mean	1.190840e+05
std	1.225272e+07
min	1.000000e+02
25%	6.534000e+03
50%	8.276000e+03
75%	1.089000e+04
max	1.508483e+09

```
#looks like 1950 does have a substantially bigger mean for lotsize.
df[df['DecadeBuilt'] == 1950][['lotSizeSqFt']].describe()
```

	lotSizeSqFt
count	8.960000e+02
mean	1.700047e+06
std	5.039471e+07
min	2.352000e+03
25%	7.361000e+03
50%	8.712000e+03
75%	1.045400e+04
max	1.508483e+09

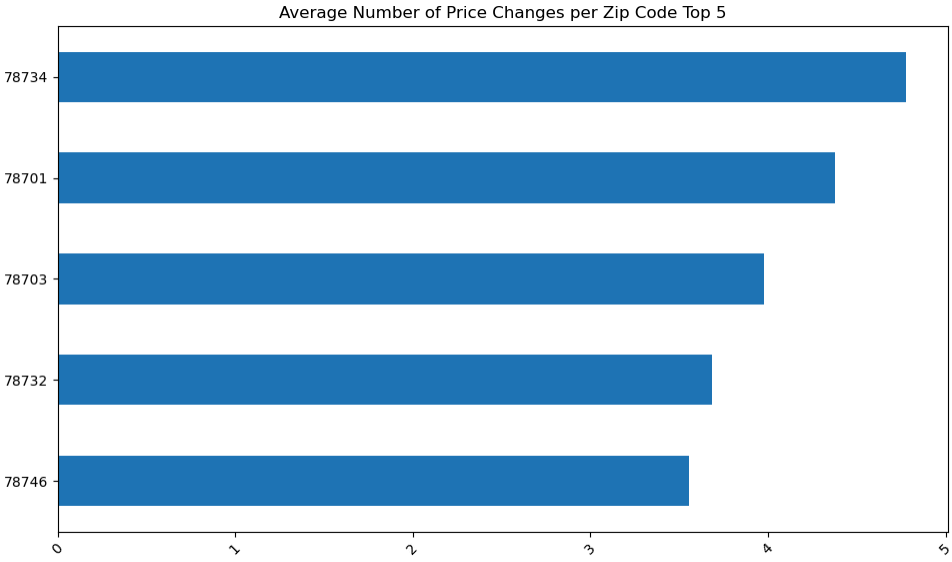
```
#once removing the largest lot size from the dataset, the lotsize average is greatly reduced
df[~(df['lotSizeSqFt'] == 1508482800.0) & (df['DecadeBuilt'] == 1950)][['lotSizeSqFt']].describe()
```

	lotSizeSqFt
count	8.950000e+02
mean	1.649038e+04
std	1.970440e+05
min	2.352000e+03
25%	7.361000e+03
50%	8.712000e+03
75%	1.045400e+04
max	5.902380e+06

```
#in fact, the decade's average lotsize becomes below the entire average when the outlier is removed from both.
df[~(df['lotSizeSqFt'] == 1508482800.0)][['lotSizeSqFt']].describe()
```

	lotSizeSqFt
count	1.517000e+04
mean	1.965327e+04
std	3.762717e+05
min	1.000000e+02
25%	6.534000e+03
50%	8.276000e+03
75%	1.089000e+04
max	3.415452e+07

Figure 6 - Price Changes By Zipcode



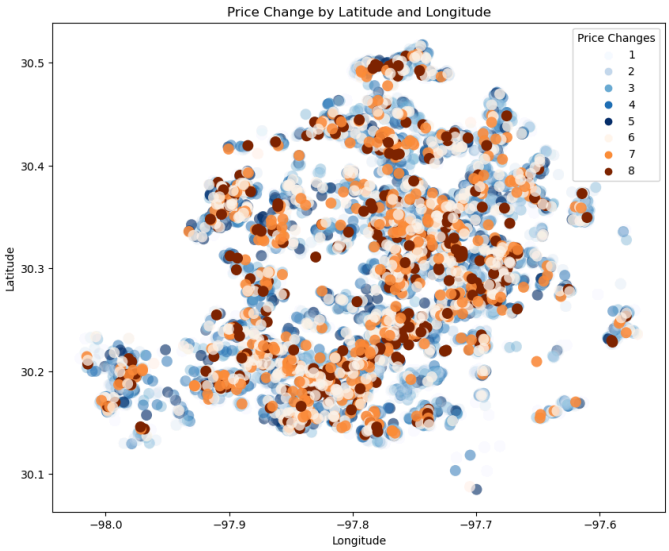


Figure 7 - Mismatch Price Outliers

5 bd | 5 ba | 4,094 sqft

3302 Blue Jay Ln, Austin, TX 78732

Off market

Zestimate®: **$1,633,900** | Rent Zestimate®: **$7,325**

Est. refi payment: $11,087/mo Refinance your loan

	city	streetAddress	zipcode	latestPrice
844	austin	3302 Blue Jay Ln	78732	5500.0

Figure 8 - Output of Grid Search and Models

Best parameters for RandomForestRegressor: {'estimator__max_depth': 15, 'estimator__min_samples_leaf': 2, 'estimator__min_samples_split': 10, 'estimator__n_estimators': 100}

Best cross-validation score (RMSE) for RandomForestRegressor: 85,615.72
Test set RMSE for RandomForestRegressor: 86,360.22
Test set R^2 for RandomForestRegressor: 0.752

Fitting 5 folds for each of 32 candidates, totalling 160 fits
Best parameters for GradientBoostingRegressor: {'estimator__learning_rate': 0.1, 'estimator__max_depth': 5, 'estimator__min_samples_leaf': 1, 'estimator__min_samples_split': 2, 'estimator__n_estimators': 200}
Best cross-validation score (RMSE) for GradientBoostingRegressor: 84,420.25
Test set RMSE for GradientBoostingRegressor: 86,986.65
Test set R^2 for GradientBoostingRegressor: 0.748

Linear Regression Test set MSE: 235,833.08
Linear Regression Test set R^2: -0.850

Figure 9.1 - Basic Statistics of the Latest Prices (Before Cleaning)

Basic Statistics of the Latest Prices

count	$15,171
mean	$512,768
std	$453,185
min	$5,500
25%	$309,000
50%	$405,000
75%	$575,000
max	$13,500,000

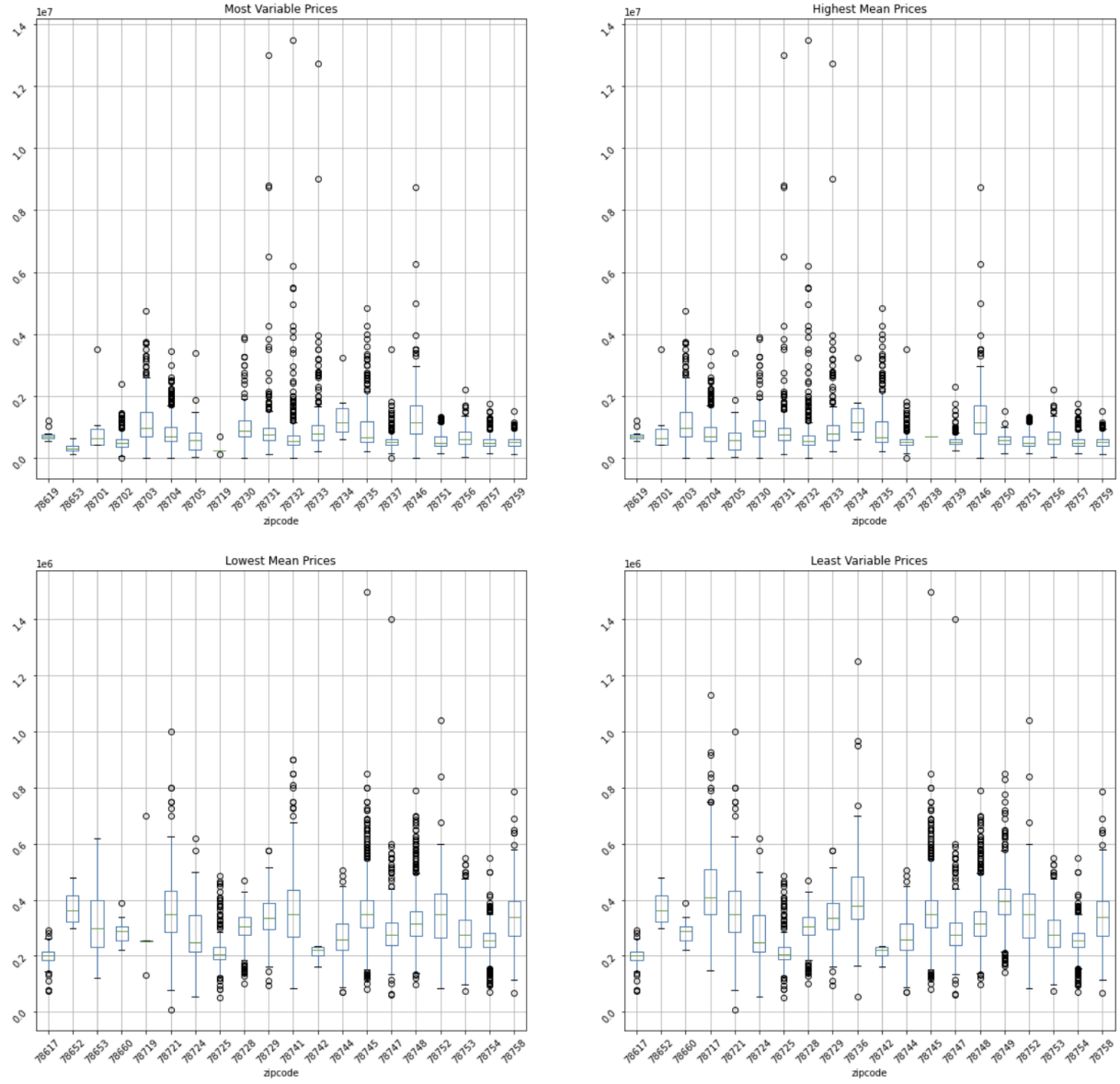
Figure 9.2 - Basic Statistics of the Latest Prices (Cleaned)

Basic Statistics of the Latest Prices (Cleaned)

count	$14,038
mean	$427,570
std	$172,580
min	$102,000
25%	$299,999
50%	$395,000
75%	$524,999
max	$970,000

Name: latestPrice, dtype: object

Figure 10




```
#interestingly, although not suprisingly, 78705 is in the top 20 most variable and highest mean for prices
print(most_variable.index.get_loc(78705) + 1) #10th most variable
print(highest_mean.index.get_loc(78705) + 1) #14th highest mean
```

```
10
14
```

Figure 11 - Top 10 Features for Models and Bottom 10 for Final Model

	Linear Regression	Random Forest	Gradient Boosting Model	Bottom 10 Features (RF)
0	avgSchoolRating	avgSchoolRating	avgSchoolRating	bedBathRatio
1	livingAreaSqFt	livingAreaSqFt	livingAreaSqFt	numOfWaterfrontFeatures
2	avgSchoolSize	latitude	latitude	numOfCommunityFeatures
3	parkingSpaces	avgSchoolSize	longitude	numOfAccessibilityFeatures
4	garageSpaces	longitude	avgSchoolSize	numOfPrimarySchools
5	DecadeBuilt	yearBuilt	yearBuilt	hasGarage
6	propertyTaxRate	lotSizeSqFt	lotSizeSqFt	hasCooling
7	hasAssociation	numPriceChanges	numPriceChanges	numOfMiddleSchools
8	storiesLivingRatio	avgSchoolDistance	MedianStudentsPerTeacher	hasView
9	numOfPrimarySchools	storiesLivingRatio	avgSchoolDistance	numOfWindowFeatures

Figure 12 - Top 5 Highest and Lowest Standard Deviations

Top 5 Highest Standard Deviations

zipcode	std_dev
78733	$1,202,543.85
78701	$1,036,326.80
78746	$1,028,027.10
78731	$904,376.08
78734	$805,865.90

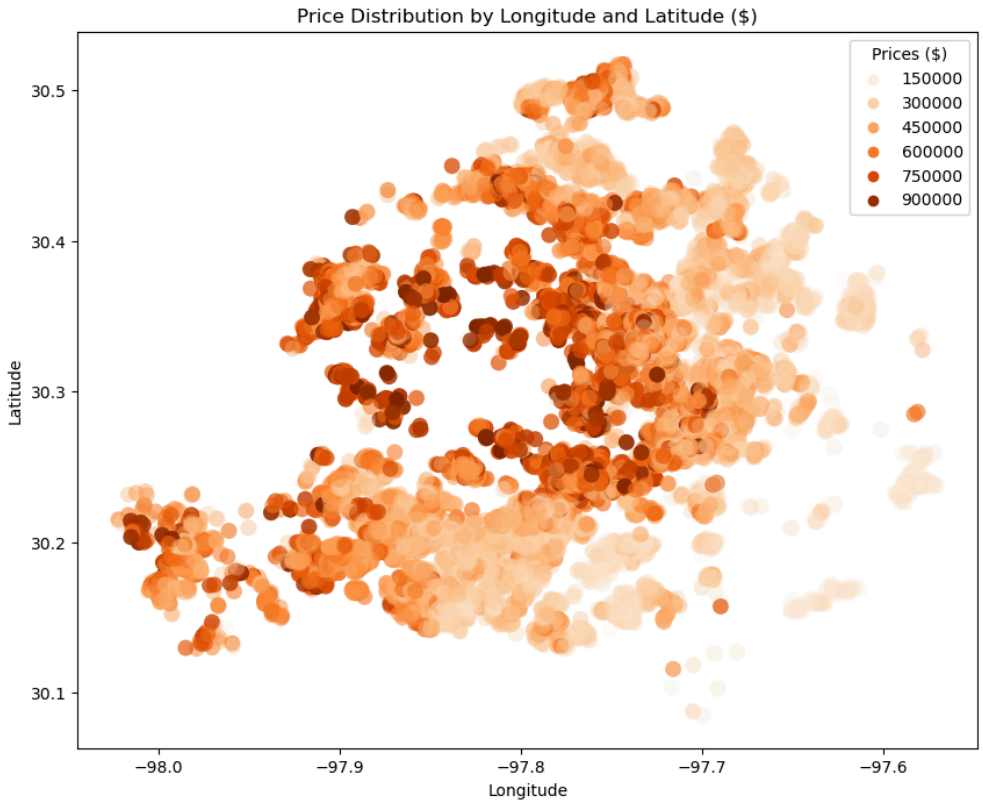
Top 5 Lowest Standard Deviations

zipcode	std_dev
78728	$60,487.05
78754	$59,504.39
78617	$37,858.22
78660	$35,478.91
78742	$34,380.26

Figure 13 - Number of Homes Sold per Zip Code



Figure 14 - Price Distribution by Zip Code



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